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Multiple channels based cooling device for chips

Abstract

A multi-channel cold plate for cooling chip wherein a first set of cooling channels function as main cooling channels and a second set of cooling channels function as a secondary and/or backup cooling channels. The two sets of cooling channels are fluidly isolated from each other, such that cooling fluid from one sent of channels cannot flow or intermix with the cooling fluid of the other cooling channel. The secondary cooling channels can be operated when demand for heat removal is increased or when the main cooling channels is unable to manage the thermal condition of the chip properly.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4470450	12/1983	Bizzell	165/104.33	F28D 15/043
9500413	12/2015	Rice	N/A	F28D 15/0233
10959352	12/2020	Chen	N/A	H05K 7/20772
11044835	12/2020	Chiu	N/A	H05K 7/20309
2009/0166008	12/2008	Lai	165/185	F28D 15/046
2012/0279686	12/2011	Chainer	165/104.21	H05K 7/20936
2016/0227672	12/2015	Lin	N/A	H05K 7/20254
2019/0203983	12/2018	Jeon	N/A	F25B 21/02
2020/0103947	12/2019	Rush	N/A	G06F 1/20
2022/0369513	12/2021	Franz	N/A	H05K 1/0201

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-03062686	12/2002	WO	F28D 15/0266

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Background/Summary

FIELD OF THE INVENTION

(1) Embodiments of the present invention relate generally to enhanced and reliable cooling of advanced microchips, such as microchips used in servers within data centers.

BACKGROUND

(2) Cooling is a prominent factor in a computer system and data center design. The number of high performance electronic components, such as high performance processors packaged inside servers, has steadily increased, thereby increasing the amount of heat generated and dissipated during the ordinary operations of the servers. The proper operation of these processors is highly dependent on reliable removal of the heat they generate. Thus, proper cooling of the processors can provide high overall system reliability.

(3) Electronics cooling is very important for computing hardware and other electronic devices, such as CPU servers, GPU servers, storage servers, networking equipment, edge and mobile system, on-vehicle computing box and so on. All these devices and computers are used for critical businesses and are the fundamentals of a company's daily business operations. The design of the hardware component and electronics packaging needs to improve to continuously support the performance requirements. Cooling of these electronic devices becomes more and more challenging to ensure that they function properly by constantly providing properly designed and reliable thermal

environments.
(4) Many advanced chips, and especially high power density chips, require liquid cooling. These chips are exceedingly expensive, so that every effort need to be taken to ensure proper heat removal from these chips. Moreover, the liquid cooling equipment must be highly reliable, since any irregularity in heat removal may lead to loss of the chips, causing loss of available processing time during the replacement operation, and even potential impact on the service level agreement which was handled by the lost chips.
(5) While liquid cooling solution must provide the required thermal performance and reliability, since data centers may have thousands of chips requiring liquid cooling, the cost of the liquid cooling system must remain acceptable. The cost of liquid cooling systems may include the cost of introducing redundancy to enhance reliability. Additionally, since different chips have different cooling requirements, it would be desirable to provide a cooling design that is adaptable to these different requirements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Embodiments of the invention are illustrated by way of example and not limitation in the figures of the accompanying drawings in which like references indicate similar elements.
(2) FIG. **1** is a block diagram illustrating an example of a data center facility according to one embodiment.
(3) FIG. **2** is a block diagram illustrating an example of an electronic rack according to one embodiment.
(4) FIG. **3** is a block diagram illustrating an example of a cold plate configuration according to one embodiment.
(5) FIG. **4** is a conceptual schematic illustrating a cross-section of a cooling plate according to an embodiment.
(6) FIG. **5** is a general schematic of a cooling plate having a corrugated structure, according to an embodiment.
(7) FIG. **6** illustrates a cooling plate wherein the secondary cooling channels are self-contained evaporation chambers, according to an embodiment.
(8) FIG. **7** is a top view of a cooling plate having heat spreaders, according to an embodiment.
(9) FIG. **8** illustrate, fluid flow and distribution in a cold plate according to disclosed embodiments.
(10) FIG. **9** illustrates a general overview of a cooling system employing a cold plate, according to an embodiment.
(11) FIG. **10** is a general schematic illustrating the operation of a cooling system according to an embodiment.
DETAILED DESCRIPTION
(12) Various embodiments and aspects of the inventions will be described with reference to details discussed below, and the accompanying drawings will illustrate the various embodiments. The following description and drawings are illustrative of the invention and are not to be construed as limiting the invention. Numerous specific details are described to provide a thorough understanding of various embodiments of the present invention. However, in certain instances, well-known or conventional details are not described in order to provide a concise discussion of embodiments of the present inventions.
(13) Reference in the specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in conjunction with the embodiment can be included in at least one embodiment of the invention. The appearances of the phrase “in one embodiment” in various places in the specification do not necessarily all refer to the same

embodiment.
(14) Disclosed embodiments provide an enhance cooling system for electronic devices, which may include single-chip module (SCM), system on a chip (SoC), multi-chip module (MCM), System in package (SIP), etc. For brevity, these are referred to herein as microchips or simply chips, but any such reference should be understood to include any of these and similar variances. As can be understood, such devices may have different cooling requirements, even within different areas of the encapsulated device itself. That is, different areas of the packaging may require different rate of heat removal.
(15) A cooling solution is disclosed which provides increased reliability and heat removal capacity. The redundant design may be used either to increase the heat removal rate on demand, or to provide backup cooling when one cooling system fails. Disclosed embodiments also enable tailoring different heat removal rates at different areas of the packaging. The following disclosure starts by providing background information regarding example application of the disclosed embodiments, and then proceeds to disclose specific embodiments. While the example relates to operation within a data center, the disclosed embodiments are not limited to data centers.
(16) FIG. **1** is a block diagram illustrating an example of a data center or data center unit according to one embodiment. In this example, FIG. **1** shows a top view of at least a portion of a data center. Referring to FIG. **1**, according to one embodiment, data center system **100** includes one or more rows of electronic racks of information technology (IT) components, equipment or instruments **101-102**, such as, for example, computer servers or computing nodes that provide data services to a variety of clients over a network (e.g., the Internet). In this embodiment, each row includes an array of electronic racks such as electronic racks **110**A-**110**N. However, more or fewer rows of electronic racks may be implemented. Typically, rows **101-102** are aligned in parallel with frontends facing towards each other and backends facing away from each other, forming aisle **103** in between to allow an administrative person walking therein. However, other configurations or arrangements may also be applied. For example, two rows of electronic racks may back to back face each other without forming an aisle in between, while their frontends face away from each other. The backends of the electronic racks may be coupled to the room cooling liquid manifolds.
(17) In one embodiment, each of the electronic racks (e.g., electronic racks **110**A-**110**N) includes a housing to house a number of IT components arranged in a stack operating therein. The electronic racks can include a cooling liquid manifold, a number of server slots (e.g., standard shelves or chassis configured with an identical or similar form factor), and a number of server chassis (also referred to as server blades or server shelves) capable of being inserted into and removed from the server slots. Each server chassis represents a computing node having one or more processors, a memory, and/or a persistent storage device (e.g., hard disk), where a computing node may include one or more servers operating therein. At least one of the processors is attached to a liquid cold plate (also referred to as a cold plate assembly) to receive cooling liquid. In addition, one or more optional cooling fans are associated with the server chassis to provide air cooling to the computing nodes contained therein. Note that the cooling system **120** may be coupled to multiple data center systems such as data center system **100**.
(18) In one embodiment, cooling system **120** includes an external liquid loop connected to a cooling tower or a dry cooler external to the building/housing container. The cooling system **120** can include, but is not limited to evaporative cooling, free air, rejection to large thermal mass, and waste heat recovery designs. Cooling system **120** may include or be coupled to a cooling liquid source that provide cooling liquid.
(19) In one embodiment, each server chassis is coupled to the cooling liquid manifold modularly, such that a server chassis can be removed from the electronic rack without affecting the operations of remaining server chassis in the electronic rack and the cooling liquid manifold. In another embodiment, each server chassis is coupled to the cooling liquid manifold through a quick-release coupling assembly having a server liquid intake connector and a server liquid outlet connector

coupled to a flexible hose to distribute the cooling liquid to the processors. The server liquid intake connector is to receive cooling liquid via a rack liquid intake connector from a cooling liquid manifold mounted on a backend of the electronic rack. The server liquid outlet connector is to emit warmer or hotter liquid carrying the heat exchanged from the processors to the cooling liquid manifold via a rack liquid outlet connector and then back to a coolant distribution unit (CDU) within the electronic rack.

(20) In one embodiment, the cooling liquid manifold disposed on the backend of each electronic rack is coupled to liquid supply line **132** (also referred to as a room supply manifold) to receive cooling liquid from cooling system **120**. The cooling liquid is distributed through a liquid distribution loop attached to a cold plate assembly on which a processor is mounted to remove heat from the processors. A cold plate is configured similar to a heat sink with a liquid distribution tube attached or embedded therein. The resulting warmer or hotter liquid carrying the heat exchanged from the processors is transmitted via liquid return line **131** (also referred to as a room return manifold) back to cooling system **120**.

(21) Liquid supply/return lines **131-132** are referred to as data center or room liquid supply/return lines (e.g., global liquid supply/return lines), which supply cooling liquid to all of the electronic racks of rows **101-102**. The liquid supply line **132** and liquid return line **131** are coupled to a heat exchanger of a CDU located within each of the electronic racks, forming a primary loop. The secondary loop of the heat exchanger is coupled to each of the server chassis in the electronic rack to deliver the cooling liquid to the cold plates of the processors.

(22) In one embodiment, data center system **100** further includes an optional airflow delivery system **135** to generate an airflow to cause the airflow to travel through the air space of the server chassis of the electronic racks to exchange heat generated by the computing nodes due to operations of the computing nodes (e.g., servers) and to exhaust the airflow exchanged heat to an external environment or a cooling system (e.g., air-to-liquid heat exchanger) to reduce the temperature of the airflow. For example, air supply system **135** generates an airflow of cool/cold air to circulate from aisle **103** through electronic racks **110**A-**110**N to carry away exchanged heat.

(23) The cool airflows enter the electronic racks through their frontends and the warm/hot airflows exit the electronic racks from their backends. The warm/hot air with exchanged heat is exhausted from room/building or cooled using a separate cooling system such as an air-to-liquid heat exchanger. Thus, the cooling system is a hybrid liquid-air cooling system, where a portion of the heat generated by a processor is removed by cooling liquid via the corresponding cold plate, while the remaining portion of the heat generated by the processor (or other electronics or processing devices) is removed by airflow cooling. Moreover, the liquid cooling may be multi-phase system wherein fluid flows in liquid or gaseous phase.

(24) FIG. **2** is block diagram illustrating an electronic rack according to one embodiment. Electronic rack **200** may represent any of the electronic racks as shown in FIG. **1**, such as, for example, electronic racks **110**A-**110**N. Referring to FIG. **2**, according to one embodiment, electronic rack **200** includes, but is not limited to, CDU **201**, rack management unit (RMU) **202**, and one or more server chassis **203**A-**203**E (collectively referred to as server chassis **203**). Server chassis **203** can be inserted into an array of server slots (e.g., standard shelves) respectively from frontend **204** or backend **205** of electronic rack **200**. Note that although there are five server chassis **203**A-**203**E shown here, more or fewer server chassis may be maintained within electronic rack **200**. Also note that the particular positions of CDU **201**, RMU **202**, and/or server chassis **203** are shown for the purpose of illustration only; other arrangements or configurations of CDU **201**, RMU **202**, and/or server chassis **203** may also be implemented. In one embodiment, electronic rack **200** can be either open to the environment or partially contained by a rack container, as long as the cooling fans can generate airflows from the frontend to the backend.

(25) In addition, for at least some of the server chassis **203**, an optional fan module (not shown) is associated with the server chassis. Each of the fan modules includes one or more cooling fans. The

fan modules may be mounted on the backends of server chassis **203** or on the electronic rack to generate airflows flowing from frontend **204**, traveling through the air space of the sever chassis **203**, and existing at backend **205** of electronic rack **200**.

(26) In one embodiment, CDU **201** mainly includes heat exchanger **211**, liquid pump **212**, and a pump controller (not shown), and some other components such as a liquid reservoir, a power supply, monitoring sensors and so on. Heat exchanger **211** may be a liquid-to-liquid or multi-phase heat exchanger. Heat exchanger **211** includes a first loop with inlet and outlet ports having a first pair of liquid connectors coupled to external liquid supply/return lines **131-132** to form a primary loop. The connectors coupled to the external liquid supply/return lines **131-132** may be disposed or mounted on backend **205** of electronic rack **200**. The liquid supply/return lines **131-132**, also referred to as room liquid supply/return lines, may be coupled to cooling system **120** as described above. As will be shown below when disclosing specific embodiments of the cold plates, the arrangement of liquid supply/return lines **131-132** may be duplicated to provide two independent cooling loops. One loop may operate continuously while the second may operate when additional cooling capacity is needed or as a backup.

(27) In addition, heat exchanger **211** further includes a second loop with two ports having a second pair of liquid connectors coupled to liquid manifold **225** (also referred to as a rack manifold) to form a secondary loop, which may include a supply manifold (also referred to as a rack liquid supply line or rack supply manifold) to supply cooling liquid to server chassis **203** and a return manifold (also referred to as a rack liquid return line or rack return manifold) to return warmer liquid back to CDU **201**. Note that CDUs **201** can be any kind of CDUs commercially available or customized ones. Thus, the details of CDUs **201** will not be described herein.

(28) Each of server chassis **203** may include one or more IT components (e.g., central processing units or CPUs, general/graphic processing units (GPUs), memory, and/or storage devices). Each IT component may perform data processing tasks, where the IT component may include software installed in a storage device, loaded into the memory, and executed by one or more processors to perform the data processing tasks. Server chassis **203** may include a host server (referred to as a host node) coupled to one or more compute servers (also referred to as computing nodes, such as CPU server and GPU server). The host server (having one or more CPUs) typically interfaces with clients over a network (e.g., Internet) to receive a request for a particular service such as storage services (e.g., cloud-based storage services such as backup and/or restoration), executing an application to perform certain operations (e.g., image processing, deep data learning algorithms or modeling, etc., as a part of a software-as-a-service or SaaS platform). In response to the request, the host server distributes the tasks to one or more of the computing nodes or compute servers (having one or more GPUs) managed by the host server. The compute servers perform the actual tasks, which may generate heat during the operations.

(29) Electronic rack **200** further includes optional RMU **202** configured to provide and manage power supplied to servers **203**, and CDU **201**. RMU **202** may be coupled to a power supply unit (not shown) to manage the power consumption of the power supply unit. The power supply unit may include the necessary circuitry (e.g., an alternating current (AC) to direct current (DC) or DC to DC power converter, battery, transformer, or regulator, etc.,) to provide power to the rest of the components of electronic rack **200**.

(30) In one embodiment, RMU **202** includes optimization module **221** and rack management controller (RMC) **222**. RMC **222** may include a monitor to monitor operating status of various components within electronic rack **200**, such as, for example, computing nodes **203**, CDU **201**, and the fan modules. Specifically, the monitor receives operating data from various sensors representing the operating environments of electronic rack **200**. For example, the monitor may receive operating data representing temperatures of the processors, cooling liquid, and airflows, which may be captured and collected via various temperature sensors. The monitor may also receive data representing the fan power and pump power generated by the fan modules and liquid

pump **212**, which may be proportional to their respective speeds. These operating data are referred to as real-time operating data. Note that the monitor may be implemented as a separate module within RMU **202**.

(31) Based on the operating data, optimization module **221** performs an optimization using a predetermined optimization function or optimization model to derive a set of optimal fan speeds for the fan modules and an optimal pump speed for liquid pump **212**, such that the total power consumption of liquid pump **212** and the fan modules reaches minimum, while the operating data associated with liquid pump **212** and cooling fans of the fan modules are within their respective designed specifications. Once the optimal pump speed and optimal fan speeds have been determined, RMC **222** configures liquid pump **212** and cooling fans of the fan modules based on the optimal pump speeds and fan speeds.

(32) As an example, based on the optimal pump speed, RMC **222** communicates with a pump controller of CDU **201** to control the speed of liquid pump **212**, which in turn controls a liquid flow rate of cooling liquid supplied to the liquid manifold **225** to be distributed to at least some of server chassis **203**. Similarly, based on the optimal fan speeds, RMC **222** communicates with each of the fan modules to control the speed of each cooling fan of the fan modules, which in turn control the airflow rates of the fan modules. Note that each of fan modules may be individually controlled with its specific optimal fan speed, and different fan modules and/or different cooling fans within the same fan module may have different optimal fan speeds.

(33) Note that the rack configuration as shown in FIG. **2** is shown and described for the purpose of illustration only; other configurations or arrangements may also be applicable. For example, CDU **201** may be an optional unit. The cold plates of server chassis **203** may be coupled to a rack manifold, which may be directly coupled to room manifolds **131**-**132** without using a CDU. Although not shown, a power supply unit may be disposed within electronic rack **200**. The power supply unit may be implemented as a standard chassis identical or similar to a sever chassis, where the power supply chassis can be inserted into any of the standard shelves, replacing any of server chassis **203**. In addition, the power supply chassis may further include a battery backup unit (BBU) to provide battery power to server chassis **203** when the main power is unavailable. The BBU may include one or more battery packages and each battery package include one or more battery cells, as well as the necessary charging and discharging circuits for charging and discharging the battery cells.

(34) FIG. **3** is a block diagram illustrating a chip cold plate configuration according to one embodiment. The chip/cold plate assembly **300** can represent any of the processors/cold plate structures of server chassis **203** as shown in FIG. **2**. Referring to FIG. **3**, chip **301** is plugged onto a socket mounted on printed circuit board (PCB) or motherboard **302** coupled to other electrical components or circuits of a data processing system or server. Chip **301** also includes a cold plate **303** attached to it, which is coupled to a rack manifold that is coupled to liquid supply line **132** and/or liquid return line **131** e.g., via blind mate connectors. A portion of the heat generated by chip **301** is removed by the cold plate **303**. The remaining portion of the heat enters into an air space underneath or above, which may be removed by an airflow generated by cooling fan **304**. Note that while a single arrow is illustrated for the supply liquid and a single arrow for the return liquid, some of the embodiments disclosed below may require two separate liquid supply lines and two separate liquid return lines.

(35) FIG. **4** is a conceptual representation of a multi-channel cold plate **403** placed on top of chip **450**, according to an embodiment, which is shown in cross-section. The body of the cold plate **403** may be made out of thermally conductive material, e.g., metal, such as copper or aluminum, or out of silicon. The cold plate **403** may be attached to the chip or pressed against the chip **450** by different packaging means, but the packaging means of attachment are not pertinent to the disclosed embodiments.

(36) The multiple channels are fabricated in a single cold plate, wherein different channels function

according to different thermal design. The channels may be distributed along the entire footprint of the of the cold plate, which may cover the entire area of the chip. In disclosed embodiments, the multiple channels may be customized for operating two different phase change coolants or two different modes of cooling, thus the main and secondary cooling channels are structured for operation according different thermal cycles. The two coolants may be independently distributed in two different cooling circulation system, for example one pumpless and one actively driven by a pump. The two cooling systems may be selected from various thermal cycles, such as pumped single or two phase cooling, pumpless single loop thermosyphon system, two phase evaporative dielectric coolant, etc., and the coolants may include water or other type of phase change coolant. In one embodiment, the design includes 3D vapor chamber as well as phase change channels (thermosiphon) coexisting in a single cooling device. The 3D vapor chamber may function differently to perform heat dissipation at different rate or different capacity from the recirculating cooling channels. In such an arrangement the 3D vapor chambers are self-contained autonomous cooling chambers, while the phase change channels are actively cooled by a cooling system which removes the vapor from the channels and deliver cooled liquid to the channels.

(37) By having two different cooling channel circulations, the cooling plate can operate in different modes. For example, one cooling circulation can be operated as the main cooling and run continuously, while the second can serve as a backup, to be operated when the main cooling fails. In another example, the main cooling can be operated continuously, while the second is operated as a booster whenever the chip experiences high processing demand and therefore generates more heat. In yet another example, the size and number of channels of each coolant type can be distributed over the area of the plate according to the design of the chip, such that areas of enhanced processing get increased cooling rate, while areas of low heat dissipation get less cooling. In a further example, the cross-section and/or the footprint of the two channels is different, for example the channels of the main cooling circulation are larger than the channels of the backup or booster channels.

(38) Returning to the example illustrated in FIG. **4**, cold plate **403** has two inlet ports, **410** and **420**. The plate also has two outlet ports, but they cannot be seen in this particular view. In this example main port **410** is the port of the main cooling system and provides cooling fluid to primary channels **412** via primary manifold **414**. The primary channels **412** may include microfins to enhance heat transfer, and the main cooling system may be a pumpless two-phase cooling system (e.g., two-phase thermosiphon system). Secondary port **420** delivers cooling fluid to the secondary channels **422** via secondary manifold **424**. The secondary channels **422** may be open channels without microfins, as illustrated, or may include fins similar to the primary channels **412**, and the secondary cooling system may be a pumped two-phase cooling system. Also, as illustrated in FIG. **4**, the secondary channels **422** are smaller, i.e., occupy less area, than the main channels **412**. The main channels **412** can operate continuously as the main cooling system of the chip. The secondary channels **422** may operate as a booster or a backup cooling system that operates only when extra cooling is needed or when the main cooling system fails.

(39) Thus, generally a cold plate is provided comprising a thermally conductive plate assembly incorporating a first set of cooling channels and a second set of cooling channels, and further having an intake port for delivering cooling fluid to the first set of cooling channels and a return port for removing vapor from the first set of cooling channels, and wherein the second set of cooling channels is fluidly isolated from the first set of cooling channels such that the cooling fluid from the first set of cooling channels is prevented from flowing into the second set of cooling channels. The second set of cooling channels may also be coupled to a cooling fluid inlet and a return outlet. Each of the cooling channels of the first set of cooling channels may include fins, while each of the cooling channels of the second set of cooling channels may or may not have fins. The cooling channels of the first set of cooling channels may be interlaced with the cooling channels of the second set of cooling channels.

(40) FIG. **5** illustrates an example of an embodiment that is structure in a corrugated format, akin to a corrugated cardboard. Cold plate **503** consists of a contact plate **534**, which is in physical contact with the chip **550**. A core unit **530** is sandwiched between the contact plate **534** and the upper plate **532**. The core unit **530** is shaped into alternate ridges and grooves, thereby forming the corrugations. One side of the corrugation forms the channels for the main cooling system, e.g., channels **512**, while the other side forms the channels for the secondary cooling system, e.g., channels **522**. Consequently, the main and secondary channels are interlaced. In the example illustrated the corrugations are even, such that the channels are all of the same size. In another example, the corrugations may be skewed, such that the channels on the main cooling are of different size of the channels of the secondary cooling.

(41) In the example illustrated in FIG. **5**, the corrugations that open towards the contact plate **534** are indicated by #1 and circulate the main cooling fluid via manifold **514**, which leads to inlet **410**. The outlet cannot be seen in this view. Conversely, the corrugations that open towards the upper plate **532** are indicated by #2 and circulate the secondary cooling fluid via manifold **524**, which leads to inlet **420**. The secondary outlet cannot be seen in this view. As shown in the callout of FIG. **5**, in one example, fins **536** are added inside the corrugations marked #1, while no fins are added to the corrugations marked #2. Conversely, in other embodiments fins **536** may be included in the corrugations marked #2 as well.

(42) By the disclosed embodiments, a thermally conductive plate assembly is provided, comprising a contact plate; an upper plate; a core section sandwiched between the contact plate and the upper plate, wherein a first set of cooling channels and a second set of cooling channels are formed within the core section; and a main cooling fluid supply manifold is formed in one of the contact plate or the upper plate, and wherein the second set of cooling channels is fluidly isolated from the first set of cooling channels such that the cooling fluid from the first set of cooling channels is prevented from flowing into the second set of cooling channels. The core section may comprise alternate ridges and grooves forming corrugations. Also, the corrugations may be skewed, such that cooling channels of the first set of cooling channels are of different size than the cooling channels of the second set of cooling channels.

(43) FIG. **6** is a cross section of another embodiment incorporating multi-channel cooling plate of two different cooling methods. This particular example is shown utilizing the corrugated structure, but the features can be incorporated in any of the structures disclosed herein or other suitable structures. In this embodiment, the main cooling channels, identified by #1, are pure evaporation channels designed for thermosiphon, while the secondary cooling channels, identified as #2, are designed for two-phase cooling, wherein both evaporation and condensation occur internally within each channel.

(44) In cold plate **603** the contact plate **634** may incorporate the inlet manifold **614**, which delivers cooling fluid to main channels **612** from inlet port **610**. Contact plate **634** physically contacts the chip **650** to remove heat therefrom. Cooling channels **612** operate as evaporation channels and the vapor is collected by outlet manifold **616**, which may be incorporated in upper plate **632**, to outlet port **618**. As shown in the callouts, the main channels may incorporate cooling fins **636**.

(45) The core unit **630** is formed as a corrugated structure, wherein one side forms the main cooling channels while the opposite side forms the secondary cooling channels. In this example, the corrugations are asymmetrical or skewed, such that the size of the main cooling channels is different from the size of the secondary channels. In this particular illustration, the secondary channels are shown larger than the main channels, but the channels may be of the same size of the main channels may be larger than the secondary channels. In the embodiment of FIG. **6**, the secondary cooling channels are designed as full two-phase channels, such that evaporation and condensation occur within the channels themselves. Therefore, no fluid supply or return ports are provided for the secondary channels, which may be self-contained sealed chambers. As shown in the dashed callout, the bottom portion of each secondary channel may include a wicking block **639**,

which may sustain cooling liquid. As the cooling liquid heats up, it evaporates and rises towards the top of the channel, where it contacts the upper plate **632** and condenses. Upon condensing the liquid returns to the wicking block **639**, as exemplified by the curved arrow. To enhance condensation a heat sink **638** contacts and cools the upper plate **632**. Heat sink **638** may be an air or liquid cooled heat sink. Therefore, in this embodiment, the bottom of each secondary channel **622** is extracting heat from the chip, and the top of each of the channels need to be liquid or air cooled to enable condensation.

(46) In an alternative embodiment, illustrated in the dash-dot callout, the secondary cooling channels are formed as 3D vapor chambers. Each of the channels is a self-contained autonomous cooling unit which includes an evaporation chamber **642** and evaporation-condensation pipes **644** attached to cooling lamellas **646**. As the liquid in the evaporation chamber **642** heats up, it vaporizes and rises in the pipes **644**. The pipes are cooled by the lamellas **646**, thereby also functioning as condensers with the lamellas functioning as fins. The condensing liquid returns to the evaporation chamber **642**. Therefore, in this embodiment cooling liquid is supplied and returned only from the primary cooling channels **612**, while no supply and return lines are provided to the secondary cooling channels **622** as they are self-contained.

(47) By this disclosure, a cold plate is provided comprising a thermally conductive plate assembly incorporating a first set of cooling channels and a second set of cooling channels, and further having an intake port for delivering cooling fluid to the first set of cooling channels and a return port for removing vapor from the first set of cooling channels, and wherein the second set of cooling channels is fluidly isolated from the first set of cooling channels such that the cooling fluid from the first set of cooling channels is prevented from flowing into the second set of cooling channels, wherein the second set of cooling channels comprises a plurality of isolated self-contained sealed chambers, each forming a two-phase evaporation and condensation chamber. Each of the isolated self-contained sealed chambers may include a wicking block.

(48) As can be appreciated, the heat removal rate and capacity of the main and secondary channels may be different. Consequently, the heat removal of the cold plate is non-uniform over the cooling plate's surface. There are applications where such non-uniform heat removal is acceptable or may be even desirable. However, there are applications where a uniform heat removal is desirable. An embodiment which enhances the uniformity of heat removal is illustrated in FIG. **7**.

(49) FIG. **7** is a top view of a cooling plate **703**. Cooling plate **703** may be of any of the embodiment illustrated herein, and has main channels **712** (which may include fins **736**) and secondary channels **722**. Additionally, each side of the cold plate **703** incorporates a heat spreader **752**, which may be, e.g., a thermally conductive plate such as copper plate. The heat spreaders **752** act to spread the heat among the two different cooling channels, thus evening out the heat over the surface of the cold plate **703**. Additionally, when the secondary channels are not operating, heat from the areas under the secondary channels can be removed by the primary channels since the heat will be conducted by the heat spreaders **752**. Conversely, if the cooling system of the primary channels fail and the secondary channels are operating in backup capacity, the heat from the areas under the non-functioning primary channels would be conducted by the heat spreaders **752** towards the secondary channels.

(50) Accordingly, a cold plate is provided comprising a thermally conductive plate assembly incorporating a first set of cooling channels and a second set of cooling channels, and further having an intake port for delivering cooling fluid to the first set of cooling channels and a return port for removing vapor from the first set of cooling channels, and wherein the second set of cooling channels is fluidly isolated from the first set of cooling channels such that the cooling fluid from the first set of cooling channels is prevented from flowing into the second set of cooling channels, and wherein the cold plate assembly further comprises at least one heat spreader in physical contact with the first set of cooling channels and the second set of cooling channels.

(51) FIG. **8** is a top view of a cold plate illustrating one method of creating cooling fluid flow in the

main and secondary channels. In this illustration, the fluid flow in the secondary channels is in counter-direction to the flow in the main channels, but the flow may also be in the same direction. Also, in some embodiments the fluid manifolds may be incorporated into the heat spreaders. The fluid flow of the main cooling channels **812** is illustrated by the dashed arrow, while the fluid flow of the secondary cooling channels **822** is shown by the dash-dot arrows. Fins **836** enhance the heat collection of the cooling fluid in the main channels **812**.

(52) FIG. **9** is a schematic illustrating an example of a cooling system utilizing the multi-channel cold plate **903** according to embodiments disclosed herein. While cold plate **903** may be constructed according to any of the embodiments disclosed herein, in the example of FIG. **9** cold plate **903** incorporates main cooling channels operating on thermosiphon two-phase pumpless thermal cycle, while the secondary cooling channels operated on a phase-change pumped thermal cycle. Of course, this is but only one example and other thermal cycles may be used.

(53) In FIG. **9**, cooling fluid is provided from condenser **970** to the inlet port **910** of the cold plate **903** via intake line **972**. Cooling fluid flowing in intake line **972** may be any liquid coolant selected from standard coolants. The cooling fluid flowing in the primary cooling channels **912** convert to vapor, and the vapor is returned from outlet **916** to condenser **970** via return line **974**. Similarly, cooling fluid is pumped by pump **988** from the secondary condenser **980** through intake lines **982** to inlet port **920** of the cold plate. The cooling fluid flows in secondary channels **922** and vaporizes. The vapor returns from outlet **923** to condenser **980** via return line **982**. By activating the pump the secondary cooling system is energized, either as a backup cooling or to enhance heat removal on demand.

(54) As can be seen, a cooling system for microchips is disclosed, comprising: a cold plate assembly incorporating a first set of cooling channels and a second set of cooling channels, wherein the second set of cooling channels is fluidly isolated from the first set of cooling channels such that the cooling fluid from the first set of cooling channels is prevented from flowing into the second set of cooling channels; a main condenser; a main supply line fluidly coupled between the main condenser and the first set of cooling channels; and a main return line fluidly coupled between the main condenser and the first set of cooling channels. The system may also include a secondary condenser; a secondary supply line fluidly coupled between the secondary condenser and the first set of cooling channels; a pump coupled to the secondary supply line; and a secondary return line fluidly coupled between the secondary condenser and the first set of cooling channels. Alternatively, the second set of cooling channels comprises a plurality of isolated self-contained sealed chambers, each forming a two-phase evaporation and condensation chamber.

(55) FIG. **10** illustrates the operation of the multi-channel system, wherein two thermal cycles are employed by two separate cooling system which operate on the same cold plate. As indicated in **10**, in this particular embodiment the main system utilizes a pumpless thermal cycle, e.g., thermosiphon circulation. The secondary system utilizes pumped phase change thermal cycle. As indicated by **12**, during normal operation the pump is off, i.e., idles, and circulation in both systems may naturally occur by convection. When enhanced or backup cooling is needed, as indicated in **14**, the pump is energized so as to mechanically force fluid flow into the secondary cooling channels. Per indication in **16**, the boiling point of the coolant in the secondary system should be lower than the boiling point of the coolant in the main cooling system.

(56) Thus, by the disclosed embodiments, a microchip cold plate is provided, having a first set of cooling fluid channels and a second set of fluid channels, the second set of fluid channels fluidly isolated from the first set of fluid channels, such that cooling fluid from the first set of cooling channels cannot enter the second set of fluid channels and cooling fluid flowing in the second set of cooling channels cannot enter into the first set of fluid channels. The cooling channels of the first set of cooling channels may interlace with the cooling channels of the second set of fluid channels.

(57) According to further embodiments, an intake manifold distributes cooling fluid among the cooling channels of the first set and a return manifold collects vapor from the cooling channels of

the first set and leads the vapor to a condenser. The cooling channels of the second set of cooling channels may be self-contained two-phase chambers, or may have cooling fluid intake and vapor outlet. When the second set of cooling channels include fluid inlet and vapor outlet it is desirable to include a pump to deliver the cooling fluid to the second set of cooling channels.
(58) According to disclosed embodiments, a microchip cooling system is provided, comprising a main condenser; a cold plate includes a plurality of main cooling channels and a plurality of secondary cooling channels, the secondary cooling channels being fluidly isolated from the main cooling channels; a main intake conduit is coupled between the main condenser and an intake port of the main cooling channels and a return conduit is coupled between the main condenser and an outlet port of the main cooling channels.
(59) In the foregoing specification, embodiments of the invention have been described with reference to specific exemplary embodiments thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope of the invention as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

Claims

1. A cooling plate for cooling microchips, comprising a thermally conductive plate assembly incorporating a first set of cooling channels and a second set of two-phase cooling channels, and further having an intake port for delivering cooling fluid to the first set of cooling channels and a return port for removing vapor from the first set of cooling channels, and wherein the second set of two-phase cooling channels is fluidly isolated from the first set of cooling channels such that the cooling fluid from the first set of cooling channels is prevented from flowing into the second set of two-phase cooling channels, wherein the first set of cooling channels and the second set of two-phase cooling channels form two different cooling channel circulations, wherein the first set of cooling channels circulation is operated as a main cooling and run continuously, and the second set of two-phase cooling channels circulation is operated as a backup, wherein the second set of two-phase cooling channels circulation is operated when the main cooling fails, wherein each of the second set of two-phase cooling channels comprises an evaporation chamber and evaporation condensation pipes attached to cooling lamellas, wherein the evaporation condensation pipes are inside the second set of two-phase cooling channels, wherein the cooling lamellas are inside the second set of two-phase cooling channels, wherein the cooling fluid in the evaporation chamber heats up, the cooling fluid vaporizes and rises in the evaporation condensation pipes, and the evaporation condensation pipes are cooled by the cooling lamellas, and wherein each of the second set of two-phase cooling channels comprises a wicking block to sustain the cooling fluid.
2. The cooling plate of claim 1, wherein the second set of two-phase cooling channels comprises a plurality of isolated self-contained sealed chambers, each forming a two-phase evaporation and condensation chamber.
3. The cooling plate of claim 2, wherein each of the isolated self-contained sealed chambers comprises the wicking block.
4. The cooling plate of claim 1, wherein the second set of two-phase cooling channels comprises a plurality of secondary cooling channels coupled to a cooling fluid inlet and a return outlet.
5. The cooling plate of claim 1, wherein each of the cooling channels of the first set of cooling channels comprises fins and each of the cooling channels of the second set of two-phase cooling channels does not have fins.
6. The cooling plate of claim 1, wherein the cooling channels of the first set of cooling channels are interlaced with the cooling channels of the second set of two-phase cooling channels.
7. The cooling plate of claim 6, wherein a cross-section of the cooling channels of the first set of cooling channels is of different size than a corresponding cross section of the cooling channels of

the second set of two-phase cooling channels.
8. The cooling plate of claim 6, wherein a footprint of the cooling channels of the first set of cooling channels is of different size than a corresponding footprint of the cooling channels of the second set of two-phase cooling channels.
9. The cooling plate of claim 1, wherein the thermally conductive plate assembly comprises a contact plate; an upper plate; a core section sandwiched between the contact plate and the upper plate, wherein the first set of cooling channels and the second set of two-phase cooling channels are formed within the core section; and a main cooling fluid supply manifold formed in one of the contact plate or the upper plate.
10. The cooling plate of claim 9, wherein the core section comprises alternate ridges and grooves forming corrugations.
11. The cooling plate of claim 10, wherein the corrugations are skewed, such that cooling channels of the first set of cooling channels are of different size than the cooling channels of the second set of two-phase cooling channels.
12. The cooling plate of claim 10, further comprising a secondary cooling fluid supply manifold formed in one of the contact plate or the upper plate; and wherein the main cooling fluid supply manifold is fluidly coupled to the first set of cooling channels and the secondary cooling fluid supply manifold is fluidly coupled to the second set of two-phase cooling channels.
13. The cooling plate of claim 1, wherein the first set of cooling channels are structured for a thermal cycle selected from pumped single phase cooling, pumped two phase cooling, pumpless single loop thermosiphon cooling, and the second set of two-phase cooling channels are structured for a thermal cycle selected from pumped single phase cooling, pumped two phase cooling, pumpless single loop thermosiphon cooling different from the thermal cycle of the first set of cooling channels.
14. The cooling plate of claim 1, wherein the second set of two-phase cooling channels include a secondary fluid different from the cooling fluid delivered to the first set of cooling channels.
15. A cooling system for cooling a microchip, comprising: a cold plate assembly incorporating a first set of cooling channels and a second set of two-phase cooling channels, wherein the second set of two-phase cooling channels is fluidly isolated from the first set of cooling channels such that the cooling fluid from the first set of cooling channels is prevented from flowing into the second set of two-phase cooling channels, wherein the first set of cooling channels and the second set of two-phase cooling channels form two different cooling channel circulations, wherein the first set of cooling channels circulation is operated as a main cooling and run continuously, and the second set of two-phase cooling channels circulation is operated as a backup, wherein the second set of two-phase cooling channels circulation is operated when the main cooling fails, wherein each of the second set of two-phase cooling channels comprises an evaporation chamber and evaporation condensation pipes attached to cooling lamellas, wherein the evaporation condensation pipes are inside the second set of two-phase cooling channels, wherein the cooling lamellas are inside the second set of two-phase cooling channels, wherein the cooling fluid in the evaporation chamber heats up, the cooling fluid vaporizes and rises in the evaporation condensation pipes, and the evaporation condensation pipes are cooled by the cooling lamellas, and wherein each of the second set of two-phase cooling channels comprises a wicking block to sustain the cooling fluid; a main condenser; a main supply line fluidly coupled between the main condenser and the first set of cooling channels; a main return line fluidly coupled between the main condenser and the first set of cooling channels.
16. The cooling system of claim 15, further comprising: a secondary condenser; a secondary supply line fluidly coupled between the secondary condenser and the first set of cooling channels; a pump coupled to the secondary supply line; and, a secondary return line fluidly coupled between the secondary condenser and the first set of cooling channels.
17. The cooling system of claim 15, the second set of two-phase cooling channels comprises a

plurality of isolated self-contained sealed chambers, each forming a two-phase evaporation and condensation chamber.

18. The cooling system of claim 15, wherein the cold plate assembly comprises a contact plate; an upper plate; a core section sandwiched between the contact plate and the upper plate; wherein the first set of cooling channels and the second set of two-phase cooling channels are formed within the core section.

19. The cooling system of claim 18, further comprising at least one intake manifold formed in one of the contact plate or the upper plate and at least one return manifold formed in one of the contact plate or the upper plate.

20. The cooling system of claim 15, wherein the cold plate assembly further comprises at least one heat spreader in physical contact with the first set of cooling channels and the second set of two-phase cooling channels.